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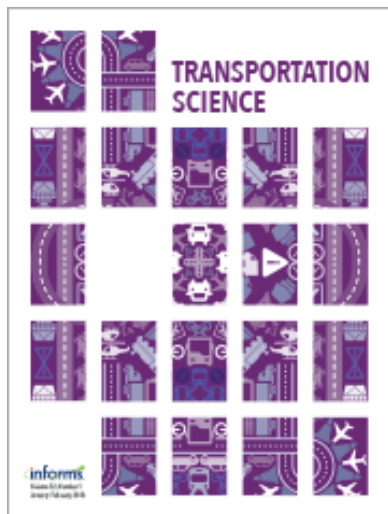
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Design, Modeling, and Analysis of Vertical Robotic Storage and Retrieval Systems

Kaveh Azadeh,^a Debjit Roy,^b René De Koster^a

^a Rotterdam School of Management, Erasmus University, 3062 PA Rotterdam, Netherlands; ^b Indian Institute of Management Ahmedabad, Ahmedabad, 380015 Gujarat, India

Contact: azadeh@rsm.nl,  <https://orcid.org/0000-0001-5267-1963> (KA); debjit@iima.ac.in,  <https://orcid.org/0000-0001-5298-9664> (DR); rkoster@rsm.nl,  <https://orcid.org/0000-0002-1740-7822> (RDK)

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Abstract. Autonomous vehicle-based storage and retrieval systems are commonly used in many fulfillment centers (e.g., e-commerce warehouses), because they allow a high- and flexible-throughput capacity. In these systems, roaming robots transport loads between a storage location and a workstation. Two main variants exist: *horizontal*, where the robots only move horizontally and use lifts for vertical transport, and a new variant *vertical*, where the robots can also travel vertically in the rack. This paper builds a framework to analyze the performance of the vertical system and compare its throughput capacity with the horizontal system. We build closed queueing network models for this that, in turn, are used to optimize the design. The results show that the optimal height-to-width ratio of a vertical system is around one. Because a large number of system robots may lead to blocking and delays, we compare the effects of different robot blocking protocols on the system throughput: robot Recirculation and Wait-on-Spot. The Wait-on-Spot policy produces a higher system throughput when the number of robots in the system is small. However, for a large number of robots in the system, the Recirculation policy dominates the Wait-on-Spot policy. Finally, we compare the operational costs of the vertical and horizontal transport systems. For systems with one load/unload (L/U) point, the vertical system always produces a similar or higher system throughput with a lower operating cost compared with the horizontal system with a discrete lift. It also outperforms the horizontal system with a continuous lift in systems with two L/U points.

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Keywords: warehousing • intralogistics • robotic technology • queueing network modeling • blocking • performance analysis

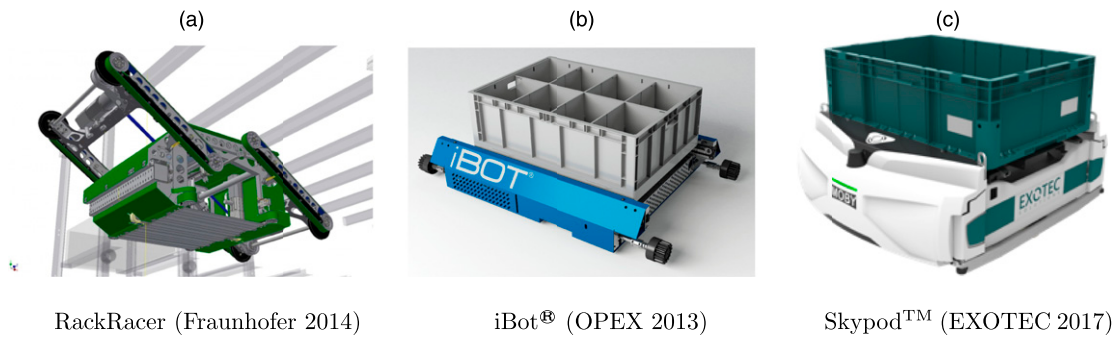
1. Introduction

The main challenge in many fulfillment centers is to adapt the picking capacity to the order volume required. The shuttle- or autonomous vehicle-based storage and retrieval system (AVS/RS) is one very popular candidate to address this challenge. In this system, a combination of autonomous shuttles and lifts is used to perform the order fulfillment process. Typically, the system throughput is constrained by the number of lifts present in this system. Shuttles move autonomously in the horizontal directions using rails and are transported in the vertical direction using lifts. Hence, we categorize them as *horizontal* systems.

Recently, robotic-based storage and retrieval systems have been developed that eliminate the multi-touch retrieval process of AVS/RSs. In these systems, a single robot can independently roam throughout the storage rack to transport items between storage locations and workstations. Two variants of robots are used in these systems. In the first group, robots

move independently in horizontal and diagonal directions to access a storage location, which we categorize as *diagonal* systems. One example of the diagonal system is a climbing robot called Rack Racer (see Figure 1(a)) developed by Fraunhofer IML (Fraunhofer 2014). In the second group, which is the focus of this study, robots move independently and autonomously in the horizontal and vertical directions inside the rack structure. Therefore, we categorize them as *vertical* systems. Perfect Pick® developed by OPEX Corporation (OPEX 2013) is one example of a vertical system using the iBot® (see Figure 1(b)) as a robot. Skypod™ (see Figure 1(c)) developed by Exotec Solutions is another example of vertical systems (EXOTEC 2017). Since 2013, vertical systems have been installed at more than 19 e-retailer warehouses, such as iHerb, BHFO, NewEgg, and Petzl in the United States, Hudson's Bay in Canada, and Cdiscount in France.

The single-touch retrieval process gives the vertical system an edge over its horizontal counterpart when

Figure 1. (Color online) Robots in Single-Touch Systems

it comes to flexibility and throughput adjustability. In a given vertical system, the desired throughput level can be obtained by only choosing the correct number of robots. However, in a given horizontal system with an already installed racking structure, the number of shuttles as well as the number of lifts need to be adjusted to achieve a certain throughput rate. Furthermore, adding additional lifts requires a major overhaul of the system. Moreover, if one of the robots breaks down in the vertical system, it can be replaced without affecting the operation. In contrast, a failure of an exchange point in the horizontal system could result in a system shutdown. Although it seems that the vertical system is more flexible and reliable compared with the horizontal system, it is not clear which system achieves a higher-throughput performance with the same storage capacity and number of robots. As a result, a comprehensive study on both systems is required to make that judgment.

Many studies exist that describe and analyze the performance of the horizontal systems (e.g., Malmberg 2002, Marchet et al. 2012, Roy et al. 2012), whereas the vertical system has not been studied yet. Therefore, we first need to investigate the vertical system before we can compare the performance and costs of the two systems. Furthermore, there are fundamental differences between the two systems that lead to a different modeling approach as well as different layout designs and control policies for the vertical system. In particular, vertical systems have roaming flexibility for robots. This provides multiple routing options to perform storage and retrieval actions, potentially leading to congestion and blocking. This can be mitigated by new block prevention policies (such as a recirculating policy,

which we present in Section 3.2) that are not possible in horizontal systems. Table 1 summarizes some of the key differences between the two systems.

The throughput capacity of the vertical system depends on several design choices, in particular the layout configuration and the number of operating robots. Although increasing the number of robots increases the system throughput capacity, it can simultaneously lead to increased blocking delays, potentially reducing the throughput capacity. Therefore, an understanding of the effect of blocking and mitigating policies on performance is crucial, especially in the conceptual design phase. To evaluate the different blocking policies and system designs, one obvious approach is to build detailed simulation models. However, developing a realistic and detailed simulation model for analyzing all possible design scenarios and parameter settings is very time consuming. Therefore, at the early conceptualization stage, analytical models are used to reduce the design search space and identify a few promising configurations, which can then be fine tuned using simulation. These analytical models are faster to evaluate, and they allow optimization by enumeration over a large number of design parameters (Zaerpour, Yu, and De Koster 2015, Tappia et al. 2016).

Hence, the objective of this paper is to answer the following research questions.

1. How do we build accurate and efficient analytical models to analyze the performance of the vertical system?
2. What is an optimal layout for the vertical system in terms of throughput performance?
3. How do the blocking delays affect the throughput performance of the vertical system?

Table 1. Comparison of the Features Between Horizontal and Vertical Systems

Category	Horizontal system (AVS/RS)	Vertical system
Physical configuration	Shuttles and lifts as storage and retrieval device; up to two L/U points per aisle (each L/U point requires a dedicated lift)	Rack-climbing robots as storage and retrieval device; up to four L/U points per aisle
Vehicle paths	Rail-based horizontal paths	Flexible horizontal and vertical paths
Load movement	Loads enter and exit a tier from one head of the aisle	Loads enter the rack section of an aisle from the top and exit from the bottom (or vice versa)
System throughput	Determined by number of shuttles and lifts	Determined by number of robots

4. Which system is better in terms of costs and throughput capacity: horizontal or vertical?

The focus and prime contribution of this paper are in the model formulation, analysis, and system comparison. The prime performance metric during the conceptualization phase is the system throughput. We build a closed queueing network to model and use the system and approximate mean value analysis (AMVA) to estimate the system throughput. By enumerating over the defined search space, we find the optimal shape of the system that maximizes the throughput rate. We analyze a *Recirculation* (REC) blocking policy and compare it with the benchmark *Wait-on-Spot* (WOS) policy for robots that are blocked. Finally, we compare the performance and operational costs of the vertical and horizontal systems.

The rest of the paper is organized as follows. In Section 2, the literature review is presented. Section 3 describes the vertical system as well as the blocking policies for the vertical system. In Section 4, we model the systems as queueing networks, and in Section 5, we present solution approaches and estimate the performance. In Section 6, the numerical analysis is performed. Specifically, we present the optimal configuration of the vertical system and compare the performance of the two blocking policies. Section 7 compares the horizontal and vertical systems for costs. Conclusions are given in Section 8.

2. Literature Review

Several studies have developed analytical models to analyze the performance of the horizontal system. The articles can be categorized into three main groups.

2.1. Modeling AVS/RSs

Malmberg (2002) was the first to analyze the horizontal (AVS/RS) system. He developed a state equation model to estimate the vehicle utilization and cycle time as a function of system design parameters (e.g., the number of shuttles, lifts, tiers, and storage columns). Malmberg (2003) extended the state equation model by including the pending transactions in the state space description as well to estimate the storage and retrieval cycle times, system utilization, and throughput capacity for various system design profiles. Note that the state equation approach is computationally inefficient for solving large-scale problems. Therefore, Kuo, Krishnamurthy, and Malmberg (2007) and Fukunari and Malmberg (2008) proposed a nested queueing approach to overcome this problem. In this approach, the lift is modeled as a queueing system in which vehicles are customers and the lifts are servers. The queue is then nested within a separate vehicle queueing system in which transactions are customers and vehicles are servers. The two systems are analyzed iteratively until the performance measures

converge. The problem with this approach is its inability to model a scenario in which the cycle starts from outside the storage rack: for example, when loads are received from outside the storage rack. To address this drawback, Fukunari and Malmberg (2009) proposed a queueing network model as an alternative to the nested queueing approach to estimate the performance of AVS/RSs. This enabled them to also capture the performance of the system when interfacing with outside systems. Although these models are efficient in estimating the vehicle utilization with reasonable accuracy, they usually fail to estimate the transaction waiting time accurately. Zhang et al. (2009) addressed this problem by dynamically choosing among three different queueing approximations based on the squared coefficient of variation of transaction interarrival times. Using this procedure, they were able to increase the accuracy of the transaction waiting time estimation significantly. Heragu et al. (2011) used an open queueing network model and an existing tool called Manufacturing Performance Analyzer to analyze the performance of the AVS/RS. Then, they provided an extensive comparison between the performances of AVS/RSs and traditional automated storage and retrieval system (AS/R) systems. Marchet et al. (2012) also used the open queueing network approach to estimate the transaction cycle time of the AVS/RS with tier-captive vehicles. Ekren et al. (2013), Cai, Heragu, and Liu (2014), and Roy et al. (2015b) used semiopen queueing networks to evaluate the performance of the AVS/RS with tier-to-tier (pooled) vehicles. Lerher et al. (2015a) derived a closed form expression of the mean travel time for single- and dual-command orders in a single-deep AVS/RS. Lerher et al. (2016) developed another closed form analytical model to estimate the throughput rate of an AVS/RS. Lerher et al. (2015b, 2017) presented a simulation model to calculate single- and dual-command cycle times and throughput of an AVS/RS.

2.2. Design Choices for AVS/RSs

Fukunari et al. (2004) studied the choice of vehicle dwell points in the AVS/RS using a decision tree analysis. Kuo, Krishnamurthy, and Malmberg (2008) used the queueing network approach to investigate the effect of the class-based storage policy on the cycle time for an AVS/RS. Ekren and Heragu (2010) used simulation in combination with a regression analysis to analyze the effect of different rack configurations on the system performance. In the regression model, they investigated the effect of three inputs (number of tiers, aisles, and bays) on system performance measures. Ekren et al. (2010) developed a simulation-based experimental design to identify several factors that affect the performance of the AVS/RS. These factors include the load/unload (L/U) point location, dwell point policy, scheduling rule, and interleaving rule

and their effect on the storage and retrieval transaction average cycle times as well as average utilization of vehicles and lifts. Roy et al. (2012) modeled one tier of the AVS/RS as a multiclass semiopen queueing network with class switching to investigate the impact of design decisions on the performance of the system. They used a decomposition approach to evaluate the effect of different system depth-to-width ratios, vehicle assignment rules, and multiple storage zones on the expected system cycle time and vehicle utilization. Marchet et al. (2013) presented a design framework for an AVS/RS with tier-captive shuttles. They used simulation to investigate the effect of rack configuration on the throughput performance of the system. Roy et al. (2015a) used a semiopen queueing network to analyze the optimal choice of dwell point location and crossaisle location. Their study showed that the end of the aisle is the optimal location for the crossaisles, whereas the L/U point dwell policy improved the performance of the system.

2.3. Control Policies for AVS/RSs

The more complex and stochastic nature of AVS/RS operation requires dynamic and real-time control policies. Especially in some variants of the system with bidirectional single lanes, there is a possibility of a deadlock in the system. He and Luo (2009) used colored time petri nets to dynamically model AVS/RSs to establish the necessary conditions to have a deadlock-free system. Roy et al. (2014) investigated the effect of vehicle blocking in the AVS/RS and proposed a semiopen queueing network to analyze the system performance and design tradeoffs. Their results showed that the blocking delays could contribute up to 20% of the transaction time. A detailed simulation model was developed by Roy et al. (2016) to evaluate the blocking delay in the AVS/RS. Carlo and Vis (2012) studied an AVS/RS with two lifts. They proposed a look-ahead strategy heuristic to simultaneously sequence the orders and assign them to each lift.

Zou et al. (2016) investigated a scenario in which the lift and vehicles in the tier-captive AVS/RS are requested simultaneously instead of sequentially. Using a fork-join queueing network, they showed that, for a system with fewer than 10 tiers, the parallel processing policy resulted in at least 5% performance improvement compared with the sequential processing policy.

The contribution of this study is threefold.

1. **Modeling.** We are the first to investigate vertical robotic-based storage systems. In these systems, robots move horizontally as well as vertically to perform order transactions. This makes them fundamentally different from previous AVS/RSs. Therefore, new models are required to capture their performance.

2. **Methodological.** We present a jump-over approximation method to analyze the REC policy. This

paper is the first to use this technique in a robotic warehousing context.

3. **Design insights.** We are the first to investigate the optimal system size of the new generation robotic-based vertical system. We also compare the effect of two blocking policies on the system throughput capacity. Furthermore, we are the first to compare cost as well as system throughput performance of vertical systems and the commonly known horizontal systems (AVS/RS).

3. System Description and Assumptions

In Section 3.1, we describe the vertical system and our modeling assumptions. In Section 3.2, we present two blocking policies for the vertical system to investigate the effect of blocking on performance. Table 2 presents the main notations used in this study.

3.1. Vertical Robotic Storage and Retrieval Systems

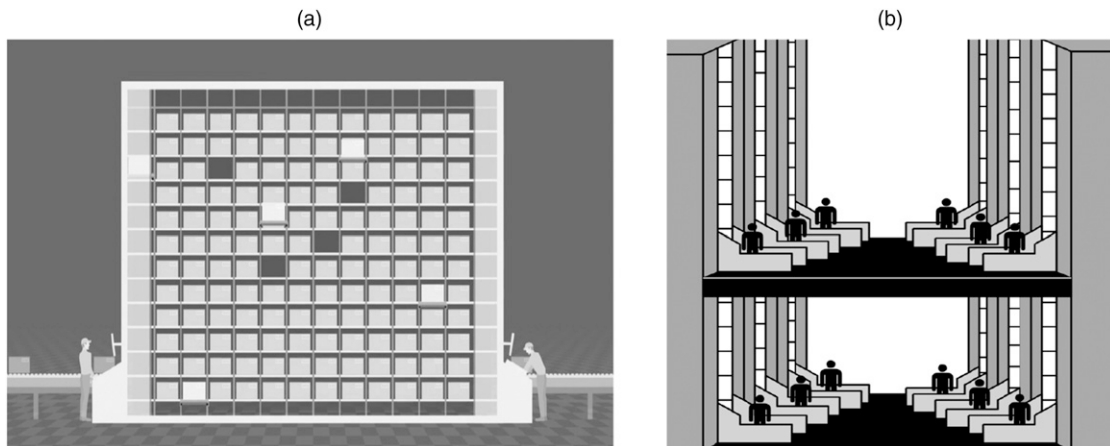
The vertical system consists of several aisles. Each aisle consists of two single-deep storage racks separated by an aisle in which autonomous robots can move. Each robot can access every storage location within the aisle by independently moving horizontally and vertically in sequence. The L/U point is located at either end or at both ends of each aisle. Additional L/U points can be included on the mezzanine floor to increase the pick capacity (see Figure 2).

Technically, the robots have the ability to move between aisles. However, in current implementations of the vertical system, the robots are captive to an aisle. Therefore, in this study, we analyze a single aisle of the vertical system. Figure 3 shows a side view of a single aisle of the vertical system with one L/U point. Each aisle is divided into several columns; we denote each column as a rack section. Although there is some level of flexibility for the robots, in recent implementations (see OPEX 2016), the robot follows a predefined direction path to access each storage location in the aisle. The predefined mostly single-directional

Table 2. Main Notations

Notation	Description
N_T	Number of tiers
N_C	Number of rack sections (columns)
n	Total number of storage positions in one aisle of the system ($2N_TN_C$)
h	Unit height clearance (tier height)
w	Unit width clearance (rack section width)
K	Number of robots (vertical)/shuttles (horizontal)
N_{LU}	Number of L/U points in the system
v_r	Velocity of the robots (vertical system)
v_s	Velocity of the shuttles (horizontal system)
v_l	Velocity of the lift (horizontal system)
τ_l, τ_u	Load and unload time
τ_{LU}	Processing time in the L/U point (we refer to this as picking time)

Figure 2. L/U Point Locations in the Vertical System



Source. (a) OPEX Corporation.

Notes. (a) Ground floor L/U point locations. (b) Mezzanine floor L/U point locations.

path minimizes congestion and avoids deadlock. The outer loop is unidirectional, whereas each rack section is bidirectional. Each time that an order is placed by a customer, a new retrieval request for a product is made.

The control system dispatches the robot from its dwell point to the requested location following the allowed direction. It picks up the tote containing the ordered product and transports it to the L/U point. A similar process is followed to store a tote back into the storage location. The travel paths corresponding to the processing sequence of the storage (Figure 4(a))

and retrieval transactions (Figure 4(b)) in a dual-command cycle are depicted.

Travel in the case of two L/U points works as follows (see Figure 5). If the dual-command cycle is initiated from the L/U point (a), the robot enters the rack section from the top and exits from the bottom. If the dual-command cycle is initiated from the L/U point (b), the robot enters the rack section from the bottom and exits from the top. Compared with L/U points with shared robots, the travel distance is smaller if the robots are dedicated to an L/U point. The reason

Figure 3. Side View of a Single Aisle

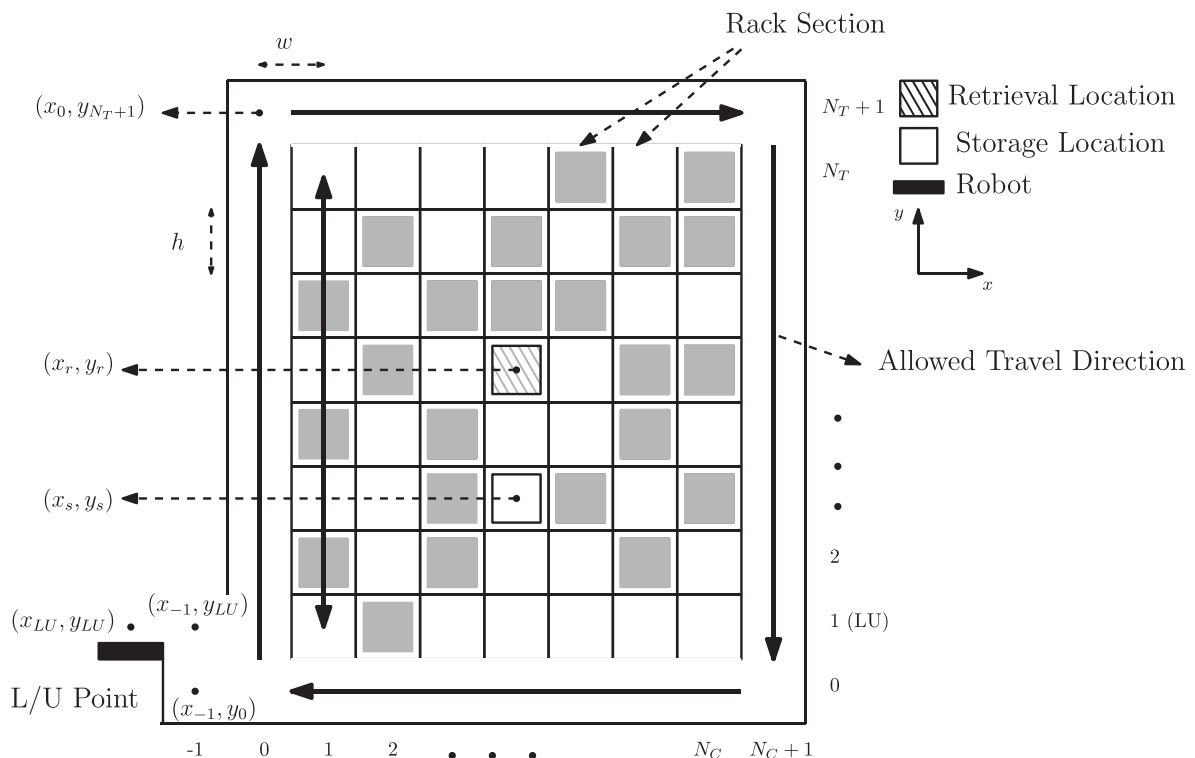
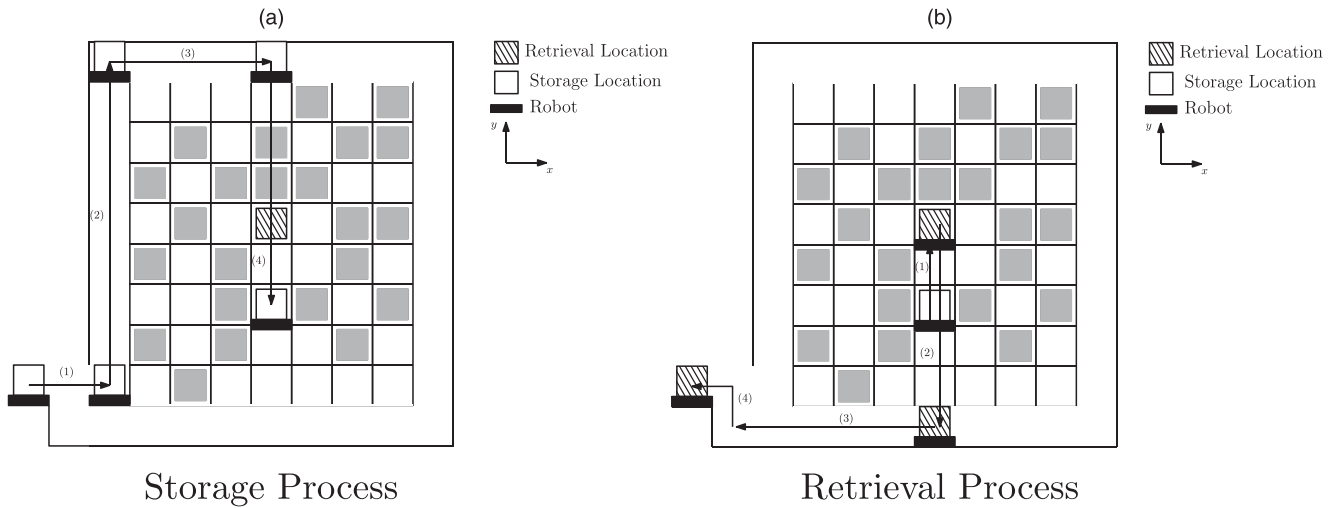


Figure 4. Dual-Command Cycle Order

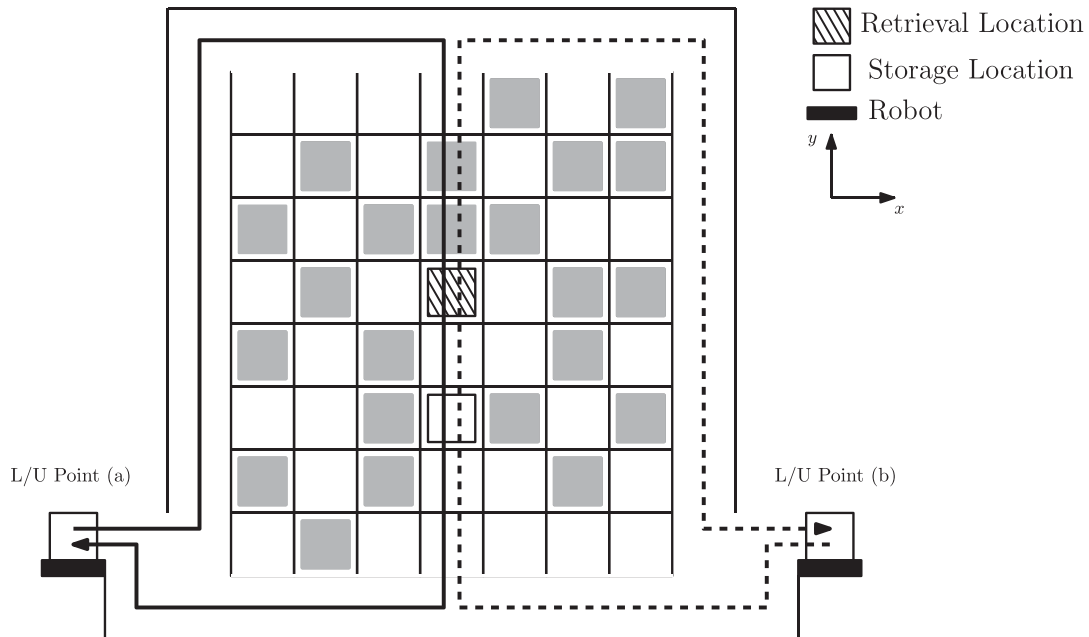
is in the predefined directed paths. If the robot starts from L/U point (a) and wants to go to L/U point (b) after picking up an item, it always passes L/U point (a) before reaching L/U point (b). Consequently, a dedicated policy is used to allocate robots to one of two L/U points, because this minimizes the travel distance. In other words, each robot serves one L/U point.

We make the following assumptions in analyzing the vertical system.

(a) Dual-command cycle orders. We assume that the robot first stores the tote containing the previously fetched items (the tote from which the order picker at the L/U point has picked some items) and then retrieves the new tote containing items for the current request. This assumption is in line with the policy that

is used in the actual implementation of the system in practice (see OPEX 2016).

(b) Same rack section. We assume that the dual-command cycle orders always belong to a specific rack section (i.e., there is always at least one storage location available in the same rack section from which the robot will retrieve the new item). Note that this assumption is not very restrictive. An average aisle with N_C rack sections and N_T tiers has $2N_CN_T$ locations. Assume that there are p empty locations. With a random storage strategy, the expected number of rack sections without empty locations is $A = N_C \left(\frac{N_C-1}{N_C} \right)^p$, which approximates to zero for large p and given N_C . Usually in automated systems, around 10% of the locations in a rack are empty. Therefore, for a typical

Figure 5. Dual-Command Cycle Travel Path in Case of Two L/U Points

rack with about 25 rack sections and 25 tiers, the number of empty locations is about $p = 125$, which is large enough to approximate A to zero.

Furthermore, storing in the closest open location within the area that has been chosen for retrieval is a common policy in the robotized systems to minimize the travel time of the robot (Nigam et al. 2014). Also, the actual implementation of the system follows the similar policy most of the time, because it reduces travel distance and increases the throughput capacity of the system. This assumption can be relaxed by routing jobs with a certain probability to other rack sections.

(c) One L/U point. We assume only one L/U point is available in each aisle unless stated otherwise (the models can be easily extended to accommodate multiple L/U points).

(d) Uniform assignment. We assume that storage and retrieval locations are assigned uniformly in the system (i.e., the probability of choosing any rack section as well as a location in any rack section is based on the uniform distribution). We further assume that the open locations in the system are independent of the retrieval requests. It is possible to accommodate different storage policies by changing the probability of accessing each rack section.

(e) One robot per rack section. We assume that only one robot can access each rack section at a certain time to avoid the risk of a deadlock in the system (because each rack section has a bidirectional path). Note that, in the actual implementation of the system, similar policy is used most of the time.

(f) Robot velocity. We incorporate acceleration and deceleration of robots and calculate the average velocity of the robot. We adopt the approach by Lerher et al. (2010) to obtain the average velocity of the robot (see Online Appendix C). We ignored the driving direction changes, because the changeover times are very short. The robots are also assumed to have

the same velocity in both horizontal and vertical directions.

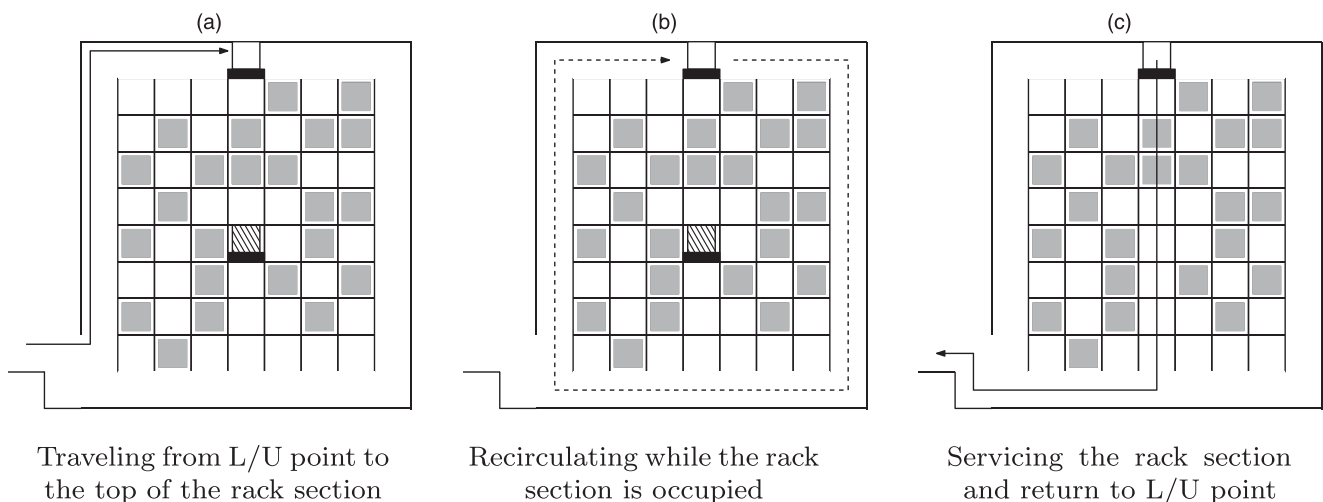
3.2. Effect of Blocking Delays on the Performance of the Vertical Robotic Storage and Retrieval Systems

To prevent a deadlock in the system, we assume that only one robot is allowed in each rack section at a certain time. If another robot wants to enter an occupied rack section, it needs to wait outside in the outer path, because there is no buffer location in a rack section for the waiting robots to queue. Consequently, increasing the number of robots can potentially lead to congestion and blocking delays in the system. We call the policy in which the blocked robot waits on top of the rack section the WOS policy. It is possible to create limited buffer locations on top of each rack section for the blocked robots to queue to ease some blocking congestion. The downside is that a higher rack is required to accommodate the buffers, which increases the robot's travel distance and potentially, the system cost.

The drawback of a WOS policy is that the waiting robot might block other robots that want to access another rack section, which negatively impacts the performance of the system. Therefore, we propose another waiting policy to mitigate this problem. In this policy (the REC policy), the robot first checks the status of the destination rack section, and if it is occupied, the robot circulates in the outer path around the rack sections. After completing one loop, it checks the status of the rack section again. If the rack section is no longer occupied, the robot claims it. Otherwise, it keeps recirculating until the rack section becomes available (Figure 6).

The REC policy can potentially result in a lower throughput, especially when the recirculation time is

Figure 6. Recirculation Policy



long. Furthermore, because the robots keep recirculating in the system, they consume more energy. However, by using this policy, we are sure that the robots do not block each other while waiting for the rack section to become empty, which can result in a higher throughput for a large number of robots.

4. Vertical System Model Description

Increasing the number of robots in the system may cause delays. However, in Section 4.1, we first discuss a model where every rack section has unlimited buffer space. This ignores the blocking effect to establish the maximum system throughput capacity. Next, we investigate the effect of blocking delays on the performance of the system. In Section 4.2, we present the model for the WOS policy, and in Section 4.3, we discuss the model for the REC policy.

4.1. Unlimited Buffer Space Inside Each Rack Section

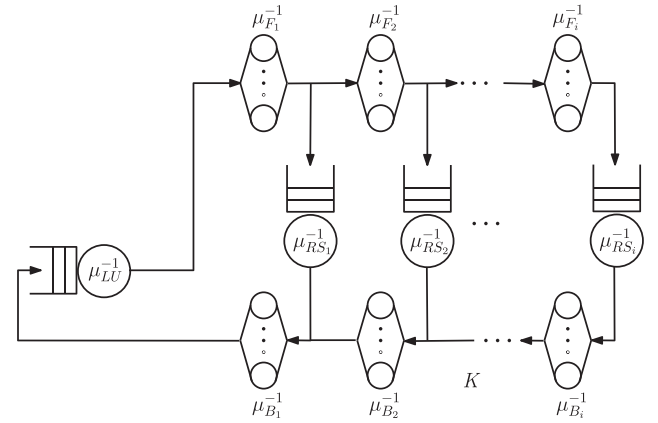
The process can be divided into three parts (see Figure 3): (1) traveling to the top of the desired rack section: loading the item at the L/U point and transporting it from (x_{LU}, y_{LU}) to (x_s, y_{N_T+1}) ; (2) the process within the rack section: descending from y_{N_T+1} to y_s , storing the tote, then going from y_s to y_r , retrieving the new tote, and finally, descending from y_r to y_0 ; (3) transporting the item back to the L/U point: going from the bottom of the rack section (x_r, y_0) to the L/U point (x_{LU}, y_{LU}) and unloading the retrieved item tote. The whole process is illustrated in Equation (1):

$$CT_{dc} = \tau_{LU} + \frac{|x_{LU} - x_0|}{v_r} + \frac{|y_{LU} - y_{N_T+1}|}{v_r} + \frac{|x_0 - x_s|}{v_r} + \frac{|y_{N_T+1} - y_s|}{v_r} + \tau_u + \frac{|y_s - y_r|}{v_r} + \tau_l + \frac{|y_r - y_0|}{v_r} + \frac{|x_r - x_0|}{v_r} + \frac{|y_0 - y_{LU}|}{v_r} + \frac{|x_0 - x_{LU}|}{v_r}. \quad (1)$$

In this section, we assume that each rack section has sufficient buffer locations for the waiting robots to queue. Figure 7 illustrates the corresponding closed queueing network used to estimate the performance of the system.

Infinite server queues with deterministic service times are used to model travel to the top of the rack section (node F_i) and travel from the rack section to the pick station (node B_i). The L/U point is modeled as a single-server queue with an exponential service time (node LU), and each rack section is modeled as a single-server queue with a generally distributed service time (node RS_i). Although there is only one L/U point in the system, the network can also accommodate a second L/U point by extending it to a multiclass closed queueing network in which robots are assigned to each L/U point based on their class.

Figure 7. Closed Queueing Network with Infinite Buffer



4.1.1. Service Time Expression. Based on the system description and the assumptions, the expected service time of each node can be calculated; μ_{LU}^{-1} depends on the speed of the picker. Equations (2) and (3) show the derivations of $\mu_{F_i}^{-1}$ and $\mu_{B_i}^{-1}$, respectively, in which $i = 1, \dots, N_C$ corresponds to the rack section number:

$$\mu_{F_i}^{-1} = \begin{cases} \frac{|x_{LU} - x_0|}{v_r} + \frac{|y_{LU} - y_{N_T+1}|}{v_r} + \frac{|x_0 - x_1|}{v_r} & \text{if } i = 1 \\ \frac{2w}{v_r} + \frac{N_T h}{v_r} + \frac{w}{v_r} & \\ \frac{|x_i - x_{i-1}|}{v_r} = \frac{w}{v_r} & \text{otherwise;} \end{cases} \quad (2)$$

$$\mu_{B_i}^{-1} = \begin{cases} \frac{|x_1 - x_{-1}|}{v_r} + \frac{|y_0 - y_{LU}|}{v_r} + \frac{|x_{-1} - x_{LU}|}{v_r} & \text{if } i = 1 \\ \frac{2w}{v_r} + \frac{h}{v_r} + \frac{w}{v_r} & \\ \frac{|x_i - x_{i-1}|}{v_r} = \frac{w}{v_r} & \text{otherwise.} \end{cases} \quad (3)$$

Equation (4) presents the travel time within RS_i . Note that y_s and y_r , the storage and retrieval locations within the rack section, respectively, are uniformly distributed:

$$T_{RS_i} = \frac{|y_{N_T+1} - y_s|}{v_r} + \tau_u + \frac{|y_s - y_r|}{v_r} + \tau_l + \frac{|y_r - y_0|}{v_r}. \quad (4)$$

The expected value and squared coefficient of variation of the service time of RS_i are

$$\mu_{RS_i}^{-1} = E[T_{RS_i}] \quad (5)$$

$$cv_{RS_i}^2 = \frac{\text{Var}[T_{RS_i}]}{[E[T_{RS_i}]]^2}, \quad (6)$$

respectively.

4.2. No Buffer Space Inside a Rack Section and WOS Blocking Policy

In the WOS policy, the robots move to the top of the rack section and check whether the rack section is occupied. If it is not occupied, it moves to the rack section. If not, it waits on top. This process can be modeled using the blocking-after-service protocol (Perros 1994), in which the robot either goes to the rack section node or waits at the previous node if the rack section is full. Figure 8 presents the corresponding queueing network.

In this network, all of the nodes are modeled as single-server queues with exponentially distributed service times (this is needed for the solution approach to solve this network, which will be discussed in Section 5.2). The L/U point is assumed to have unlimited buffer space, whereas all of the other travel nodes do not have any buffer space. The rack sections can also have a limited number of robot buffer locations. Using an unlimited buffer at the L/U point is justified, because in practice, the robots can wait on a conveyor with ample space. The description of each node as well as its expected service time are presented in the next part.

4.2.1. Service Times Expressions. The expected service times in the L/U point and the rack section do not change when including blocking in the system; therefore, μ_{LU}^{-1} and $\mu_{RS_i}^{-1}$ have the same value as the unlimited buffer network presented in Section 4.1. The travel time to move horizontally from the L/U point to the location (x_0, y_1) in Figure 3 is denoted by $\mu_{f_0}^{(-1)}$, with expectation $2w/v_r$. Furthermore, $\mu_{t_i}^{-1}$ is the travel time to climb one tier, with expectation h/v_r . Similarly, $\mu_{f_i}^{-1}$

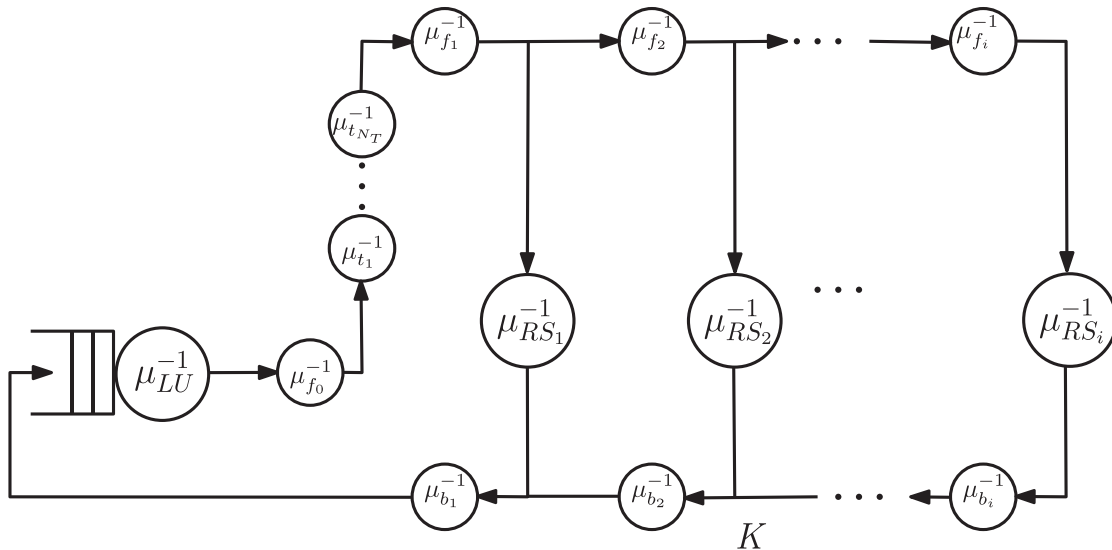
and $\mu_{b_i}^{-1}$ are the travel times to move horizontally the width of one rack section, with expectation w/v_r ; note that $i = 1, \dots, N_C$.

4.3. No Buffer Space Inside a Rack Section and REC Blocking Policy

First, we build a network for one rack section, and second, we extend it to the whole system. On service completion, the robot goes from the workstation to the top of the rack section. This process is defined by two infinite server queues F and UP . If the rack section is not occupied, the robot enters the rack section. Otherwise, it recirculates in the system by going to the infinite server queue D , which in combination with node UP , creates the outer recirculation loop. Note that the robot recirculates in the outer loop of the system (Figure 6(b)). After completing one loop, the robot checks the rack section again. Let p_b be the probability that the rack section is occupied (p_b is the same as the marginal probability of having one robot in RS). Then, with probability p_b , the robot goes to node D , and with probability $1 - p_b$, the robot enters node RS . On service completion in RS , the robot continues its route to the L/U point via the infinite server queue G , which represents the travel time between the rack section and the L/U point. Figure 9 illustrates the resulting closed queueing network.

Because the waiting robots circulate in the outer loop without stopping anywhere, we assume that no blocking occurs in the outer loop. Therefore, we can treat each rack section separately and extend the previous network to accommodate the rest of the rack sections. Figure 10 presents the resulting network.

Figure 8. Closed Queueing Network with Finite Buffer (WOS Policy)



and $\mu_{D_i}^{-1}$ are presented in Equations (7), (8), (9), and (10), respectively; note that $i = 1, \dots, N_C$:

$$\mu_{D_i}^{-1} = \frac{((N_C + 1) - i)w}{v_r} + \frac{(N_T + 1)h}{v_r} + \frac{(N_C + 1)w}{v_r} + \frac{h}{v_r}. \quad (10)$$

In this section, we provide three solution approaches to solve the networks that are presented in Section 4. Section 5.1 offers a solution to solve the network with unlimited buffer locations. Section 5.2 discusses the solution approach for solving the network for the WOS policy, and Section 5.3 presents a solution algorithm to solve the network corresponding to the REC policy.

The RS_i nodes corresponding to the closed queueing network described in Section 4.1 have generally distributed service times. Therefore, the networks do not have a product form solution (Baskett et al. 1975), and an approximation technique is required. We use

AMVA to approximate the performance of the networks. The AMVA method is an extension of the regular mean value analysis (MVA) method (Reiser and Lavenberg 1980), in which the squared coefficient of variation of the service time is used as a correction factor in calculating the residence times. The method is explained in Online Appendix B.

5.1.1. Throughput Validation with Real System. To validate the model for the vertical system, we compare the result with the output data provided by Bastian Solutions, a consulting company that has implemented such systems in practice. Table 3 presents the system parameters.

The company only reported the maximum velocity of robots without accounting for acceleration/deceleration to be 1.89 m/s. We use the approach by Lerher et al. (2010) to calculate the average velocity of the robot (see Online Appendix C). We assume that the robot has an acceleration/deceleration of 1 m/s² and calculate the average velocity of the robot inside the rack section. The resulting average velocity of the robot is 1 m/s, which we use in the rest of this study. To estimate w and h , we have added 5% to the tote dimensions as reported by the company to account for the rack size and other clearances in the rack structure. The system also has two L/U points: one at each end of the aisle. Using the given parameters, the system achieves a throughput of 800 dual-command cycles per hour according to the company. Using our analytical model, we estimate the system throughput to be 824 dual-command cycles per hour (about 3% higher than what is achieved). We also build a simulation model using the parameters in Table 3 and compare the results with our analytical model performance. The errors, which are reported in Table 4, are negligible.

5.2. Closed Queueing Network Without Buffer Location—WOS Policy

Because of the finite buffers, the queueing network developed in Section 4.2 does not have a product form solution (Perros 1994), and an approximation is needed to obtain the performance statistics. We use the approximation proposed by Akyildiz (1988), because it fits our network's characteristics. The approximation is a modified MVA for calculating the mean residence time of the jobs in the blocked node. The number of jobs in the destination node is calculated using MVA. If the calculated number of jobs is bigger than the

Table 4. Performance Statistics

	Simulation	Analytical	Error, %
Utilization rack Section 1	0.5708 ± 0.0146	0.5674	0.6
Utilization L/U point	0.1714 ± 0.016	0.1698	0.93
System throughput	822.8346 ± 1.2519	824	0.14

number of buffer spaces in the destination node, the new job is blocked. Therefore, the new job remains in the source node until a spot in the destination node becomes available. When a job finishes service at the destination node, the spot becomes available for the blocked job. Consequently, the expected residence time of the new jobs in the source node increases by the expected remaining service time of the job in the destination node. The expected remaining service time of the job in the destination node is approximated to be the same as the expected service time of the destination node. The detailed explanation of the algorithm and its accuracy evaluation can be found in Akyildiz (1988).

5.3. Closed Queueing Network Without Buffer Location—REC Policy

The queueing network developed in Section 4.3 does not have a product form solution either. To accurately estimate the performance statistics, we approximate the network by another one with the jump-over blocking protocol (Van Dijk 1988). The jump-over network has a product form solution, and its performance statistics can be calculated using MVA (Van der Gaast et al. 2018). We first present how the jump-over network is built, and then, we illustrate the MVA-based algorithm to solve the network. Table 5 presents the notations used in explaining the jump-over network and the solution algorithm.

5.3.1. Jump-Over Network. Assume that a robot that intends to visit RS_i is labeled as either *entered* RS_i or *skipped* RS_i . Technically, this labeling is done based on the robot actually visiting the RS_i (i.e., it is labeled entered RS_i if it actually entered RS_i and served, and it is labeled skipped RS_i if it skipped RS_i , because the rack section was occupied). Alternatively, we can say that the robot always enters RS_i but incurs zero service time if it is labeled skipped. However, in the jump-over network, the labeling is done randomly and regardless of whether the robot actually entered RS_i and received service. The probability of a robot receiving either one of the labels is taken as the fraction of robots in the original network receiving the specific label (i.e., the fraction of robots that are labeled as skipped RS_i equals the blocking probability of RS_i in the original network: p_{b_i}). Initially, p_{b_i} is not known; however, it can be estimated iteratively using any initial guess. Hence, it is assumed that p_{b_i} is known beforehand in the jump-over network. Therefore, P_{RS_i, D_i}

Table 3. System Parameters (Bastian Solutions 2016)

$n/2$	N_T	N_C	w , cm	h , cm	K	v_r , m/s	τ_l, τ_u , s	τ_{LU} , s
875	25	35	80	32	15	1	1.5	5

Table 5. Notations for Jump-Over Network

Notation	Description
k	Number of robots in the network
K	Total number of robots in the system
j, m, n	Nodes in the network: $LU, F, UP_i, RS_i, D_i, B_i$
i	Rack section index: $1, 2, \dots, N_C$
$T_j(k)$	Mean residence time of node j when there are k jobs in the network
$L_j(k)$	Mean number of robots at node j when there are k jobs in the network
$X(k)$	Overall throughput of the system when there are k jobs in the network
$P_{m,n}$	Routing probability of going from node m to node n
p_{b_i}	Blocking probability of node RS_i
$U_j(k)$	Utilization of node j when there are k jobs in the network
$\pi_j(q k)$	Marginal probability of having q jobs in node j when there are k jobs in the system
V_j	Visit ratio for node j
μ_j^{-1}	Expected service time of node j

(routing probability to go from RS_i to D_i) equals p_{b_i} , and P_{RS_i, G_i} (routing probability to go from RS_i to G_i) equals $1 - p_{b_i}$. This means that the robot is labeled skipped RS_i and routed to the node D_i with probability p_{b_i} , and it is labeled entered RS_i and routed to the node G_i with probability $1 - p_{b_i}$. Finally, after the robot comes out of the RS_i , it follows the mentioned Markovian route regardless of whether it skipped or served inside RS_i . Figure 11 presents the corresponding network with the jump-over protocol.

5.3.2. Visit Ratios. The visit ratios (V_j) can be calculated for each node according to the routing probabilities.

We illustrate the approach to obtain visit ratios with one rack section in the system. Then, using the same idea, the visit ratio of the system with more than one rack section can be obtained. For that purpose, we follow a robot doing a dual-command cycle route starting from the L/U point and coming back to the L/U point. Depending on how many times the robot has jumped the rack section, the number of times that it visited each node in one route is different. Table 6 presents the number of times that the robot visits each node on its route for different scenarios along with the probability of a scenario.

The number of times that the robot visits LU, F , and G in a dual-command cycle is always one. Therefore, the expected number of visits for LU, F , and G is 1 ($V_{LU} = 1, V_F = 1, V_G = 1$). Note that the robot always visits a RS even if it is blocked, but the incurred service time is zero. It is clear from Table 6 that the number of times that the robot visits UP or RS follows a geometric distribution with parameter $1 - p_b$. Therefore, the expected number of visits for UP and RS is $V_{UP} = V_{RS} = 1/(1 - p_b)$. The number of times that the robot visits D also follows the same geometric distribution but starts from zero. Hence, the visit ratio for D is $V_D = p_b/(1 - p_b)$. Similarly, the visit ratios for a network with more than one rack section can be obtained by multiplying them with the probability that the RS_i is selected.

5.3.3. Mean Value Analysis for Jump-Over Network. The idea is to initialize the p_{b_i} with any arbitrary value, solve the jump-over network using MVA, update the blocking probabilities, and repeat the MVA again until the difference between the current calculated

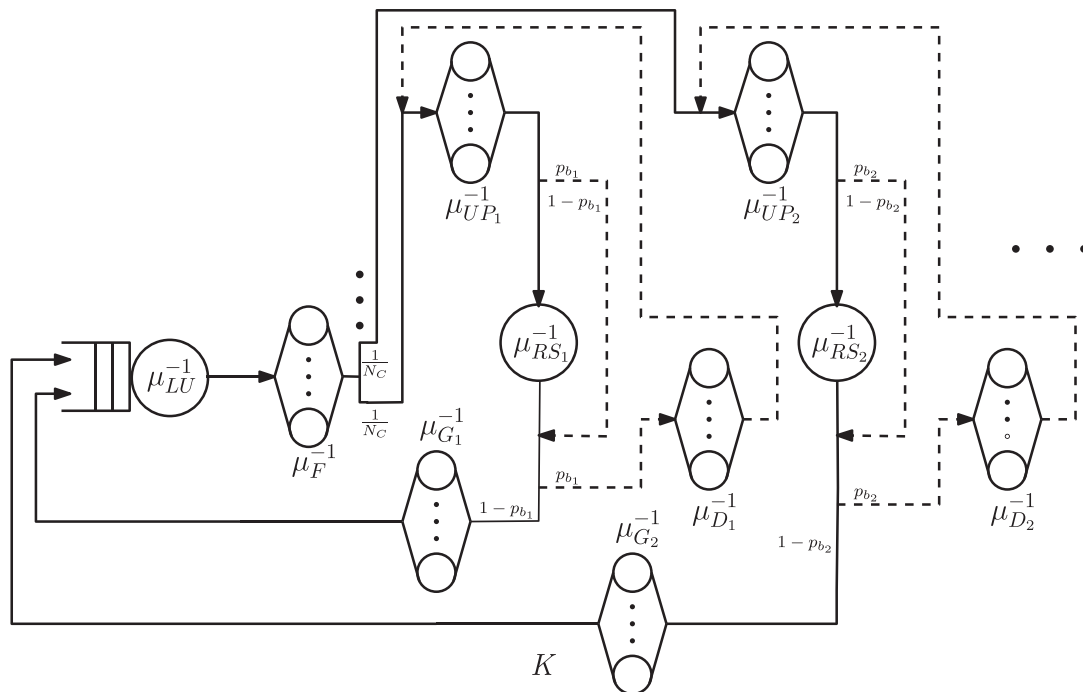
Figure 11. Jump-Over Approximation of the Closed Queueing Network in Figure 10

Table 6. Number of Times That the Robot Visits Each Node

LU	F	UP	RS	D	G	Probability
1	1	1	1	0	1	$1 - p_b$
1	1	2	2	1	1	$(1 - p_b)p_b$
1	1	3	3	2	1	$(1 - p_b)p_b^2$
1	1	4	4	3	1	$(1 - p_b)p_b^3$

blocking probability and the previous one is less than a given ϵ .

In each iteration of the MVA, given k robots in the system, the algorithm calculates $T_j(k)$, $X(k)$, $L_j(k)$, and $\pi_{RS_i}(1|k)$ (the marginal probabilities that RS_i is occupied). By iteratively increasing the number of robots in the system from zero to K , we can calculate all of the performance statistics. For the initialization phase ($k = 0$), we have $L_i(0) = 0$, $\pi_{RS_i}(0|0) = 1$, $\pi_{RS_i}(1|0) = 0$. Using the arrival theorem, the mean residence times of the LU , F , UP_i , D_i , and G_i are calculated as

$$T_j(k) = \begin{cases} \frac{1}{\mu_j}(1 + L_j(k-1)) & \text{if } j = LU \\ \frac{1}{\mu_j} & \text{if } j = F, UP_i, D_i, G_i. \end{cases} \quad (11)$$

The mean residence time of the RS_i is calculated by

$$T_{RS_i}(k) = p_{b_i}(0) + (1 - p_{b_i}) \frac{1}{\mu_{RS_i}}; \quad (12)$$

p_{b_i} is the probability that the RS_i is occupied (i.e., the probability of having one robot in RS_i when we have $k - 1$ robots in the system):

$$p_{b_i} = \pi_{RS_i}(1|k-1). \quad (13)$$

As a result,

$$T_{RS_i}(k) = \pi_{RS_i}(1|k-1) \cdot 0 + (1 - \pi_{RS_i}(1|k-1)) \frac{1}{\mu_{RS_i}}. \quad (14)$$

After the $T_j(k)$ values are obtained, the system throughput $X(k)$ is calculated by using Little's Law:

$$X(k) = \frac{k}{\sum_j V_j T_j(k)}. \quad (15)$$

Using the throughput of the system and Little's Law again, the new number of robots in each node for the next iteration is calculated:

$$L_j(k) = V_j X(k) T_j(k).$$

The marginal probabilities of having q robots in RS_i are calculated by balancing the number of transactions per time unit between state $q - 1$ and q . The rate from q to $q - 1$ is given by $\mu_{RS_i} \pi_{RS_i}(q|k)$, and by using the arrival theorem, the rate of going from $q - 1$

to q is $V_{RS_i} X(k) \pi_{RS_i}(q - 1|k - 1)$. Because q is either one or zero for RS_i , we have

$$\pi_{RS_i}(1|k) = \frac{V_{RS_i} X(k)}{\mu_{RS_i}} \pi_{RS_i}(0|k - 1). \quad (16)$$

Furthermore, from the normalization constraint, we have

$$\pi_{RS_i}(0|k) = 1 - \pi_{RS_i}(1|k). \quad (17)$$

Therefore, the marginal probabilities can be calculated by exploiting Equations (16) and (17).

The performance statistics are estimated by sequentially applying the above equations. However, before the procedure starts, the blocking probability needs to be initialized. The initial value for the blocking probabilities can be any arbitrary number between zero and one. Based on this initial value for the blocking probabilities and by using the MVA method, the marginal probability of finding RS_i containing one robot is obtained. Then, we use this value as a new initial value for the blocking probabilities and repeat the MVA again:

$$p_{b_i}^{new} = \pi_{RS_i}(1|K). \quad (18)$$

We continue the procedure until the difference between the newly calculated blocking probabilities and the current one is less than ϵ :

$$|p_{b_i}^{new} - p_{b_i}^{current}| < \epsilon.$$

In our experiments, convergence is reached fast (within a second) regardless of the initial values of p_{b_i} . The total procedure is presented in Online Appendix D.

5.3.4. Validation. We use simulation to see whether the jump-over approximation accurately estimates the performance statistics of the system with the REC policy. Instead of simulating the queueing network, we simulate a realistic implementation of an actual system to validate the model. The detailed simulation is built using the parameters listed in Table 7.

The simulation is built in AutoModTM software version 12.3, simulation software commonly used by many material handling companies. The model uses the actual REC blocking policy; the exact physical dimensions of racks, totes, and robots; and exact travel times. Only the picking process is assumed to have an exponential distribution with unlimited buffer capacity. We run the simulation 40 times and record the results with 90% confidence intervals. Then, using the same parameters, we create the jump-over approximation of the system and solve it using the proposed algorithm. The blocking probabilities of each rack section and the system throughput of the system obtained by simulation and the analytical model are compared with two levels for the number of robots K in the system: 5 and 10. Table 8 illustrates the results.

Table 7. System Parameters

$n/2$	N_T	N_C	w , cm	h , cm	K	v_r , m/s	τ_l, τ_u , s	τ_{LU} , s
8	4	2	80	32	15	1	1.5	3.6

The jump-over approximation results in less than 4% error, and it is, therefore, an accurate representation of the original system with the REC blocking policy.

Furthermore, Figure 12 illustrates the number of iterations required by the algorithm to converge for different initial values of p_{b_1} . The ϵ is set to 0.0001.

6. Numerical Analysis

Choosing the right dimensions of the system directly influences the performance of the system. In Section 6.1, we investigate different layout configurations of the vertical system using the model developed in Section 4.1 to obtain the optimal layout. Later, in Section 6.2, we compare the performance of the two blocking policies modeled in Sections 4.2 and 4.3 to evaluate under which conditions the blocking policies result in better system performance.

6.1. Optimal Rack Layout Configuration

Using the closed queueing network developed in Section 4.1 (i.e., no blocking is assumed), we analyze different layout configurations with varying height-to-width ratio (i.e., N_T/N_C) to obtain the optimal layout configuration. We use discrete optimization where we enumerate over possible parameter values in the design search space to find the optimal value. We formulate the optimization problem as follows:

$$\begin{aligned}
 & \arg \max_{N_T, N_C} X(n, K, N_T, N_C) \\
 & \text{s.t.:} \\
 & N_T N_C = \frac{n}{2} \\
 & LB_{N_T} \leq N_T \\
 & N_T \leq UB_{N_T} \\
 & LB_{N_C} \leq N_C \\
 & N_C \leq UB_{N_C} \\
 & K = \text{constant} \\
 & \frac{n}{2} = \text{constant} \\
 & N_T, N_C \in \mathbb{Z}^+.
 \end{aligned} \tag{19}$$

We investigate four levels of the number of storage locations ($n/2 = 300, 600, 900, 1,200$) and two levels of the number of robots in the system ($K = 5$ and 10). The rest of the system parameters are the same as shown in Table 3. In each scenario, the system throughput is calculated while changing the number of tiers and rack sections for their lower bound (LB) to their upper bound (UB) in a way that the total number of storage locations remains the same. The layout resulting in the maximum system throughput is identified as the optimal layout. Figure 13 illustrates system throughput for different scenarios. The x axis is the value of N_T/N_C in a logarithmic scale, and the y axis is the system throughput measured in the number of dual-command cycles per hour.

Table 9 presents the layout configuration that achieves the maximum system throughput.

Because N_T and N_C take integer values, only certain ratios of N_T/N_C can be realized for a given number of storage locations. As a result, the presented optimal N_T/N_C ratios are not identical for layouts with a different number of total storage locations. However, Figure 13 and Table 9 show that the optimal N_T/N_C ratios are close to one in each scenario.

Instead of using a queueing network, some studies adopt a probabilistic travel time approach to obtain the optimal layout configuration. In that method, the system configuration is chosen to minimize the expected dual-command travel time. In Online Appendix F, this approach is presented for the vertical system. The results are presented in Table 10. The reported numbers are the feasible values for N_T and N_C , whereas the numbers in parentheses are the actual numbers calculated by the method.

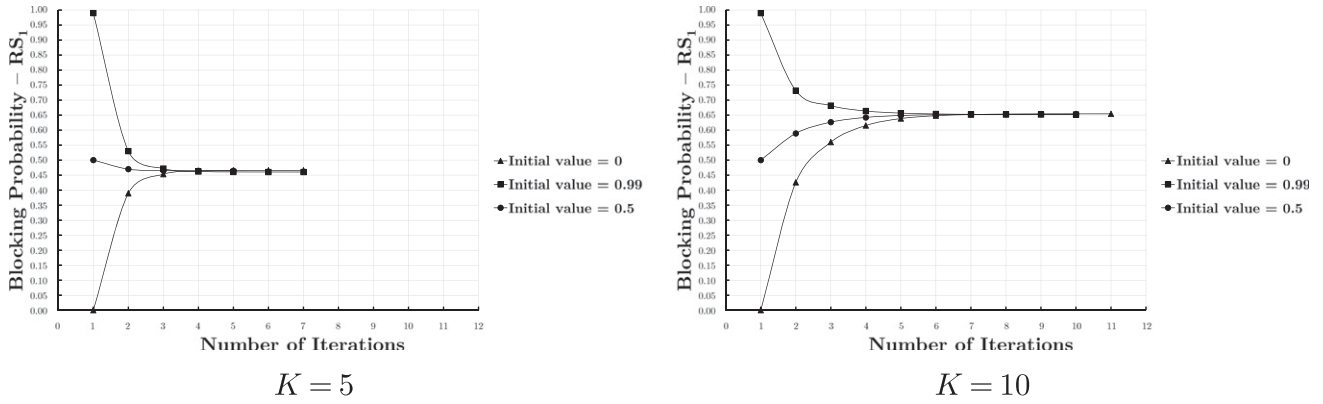
6.1.1. Analysis and Obtained Insights. Bozer and White (1984) show that, for a random storage assignment, the square-in-time rack is optimal for an automated storage and retrieval system (i.e., the expected horizontal travel time is equal to the expected vertical travel time in a single- or dual-command cycle). Our travel time analysis also provides similar results for the vertical system. Furthermore, the optimal layout resulting from the travel time expression approach leads to almost the same system throughput as the queueing network. Hence, in this particular case, we can claim that the waiting times in the L/U point and

Table 8. Performance Statistics

K	Blocking probability RS_1			Blocking probability RS_2			System throughput (DC/h)		
	Simulation	Jump	Error, %	Simulation	Jump	Error, %	Simulation	Jump	Error, %
5	0.4791 ± 0.0091	0.4626	3.44	0.4535 ± 0.0132	0.4657	2.69	683.12 ± 4.30	687.49	0.64
10	0.6393 ± 0.0081	0.6543	2.34	0.6787 ± 0.0053	0.6582	3.02	866.51 ± 6.41	895.98	3.40

Note. DC/h , dual-command cycles per hour.

Figure 12. Number of Iterations That the Algorithm Needs to Converge



rack section queues in the network do not affect the optimal layout configuration.

6.2. Comparing the Blocking Policies

Two blocking policies were modeled in Sections 4.2 and 4.3. In this section, we aim to find out how the REC policy performs compared with the basic WOS policy. We investigate the WOS policy when there is no buffer location as well as when there is one buffer location inside each rack section (WSO-0 and WSO-1,

respectively). We analyze the performance of a system with four rack sections and eight tiers ($N_C = 4$, $N_T = 8$). For the WOS-1 policy, the system has nine tiers ($N_T = 9$) to accommodate the buffer location. To see the blocking effect on performance, the system needs to be the bottleneck and not the picker. Therefore, we assume that we have four servers (pickers) at the single L/U point each with a picking time of 3.6 seconds (which corresponds to the current state-of-the-art picking stations that achieve 1,000 dual-command

Figure 13. System Throughput in Different Scenarios

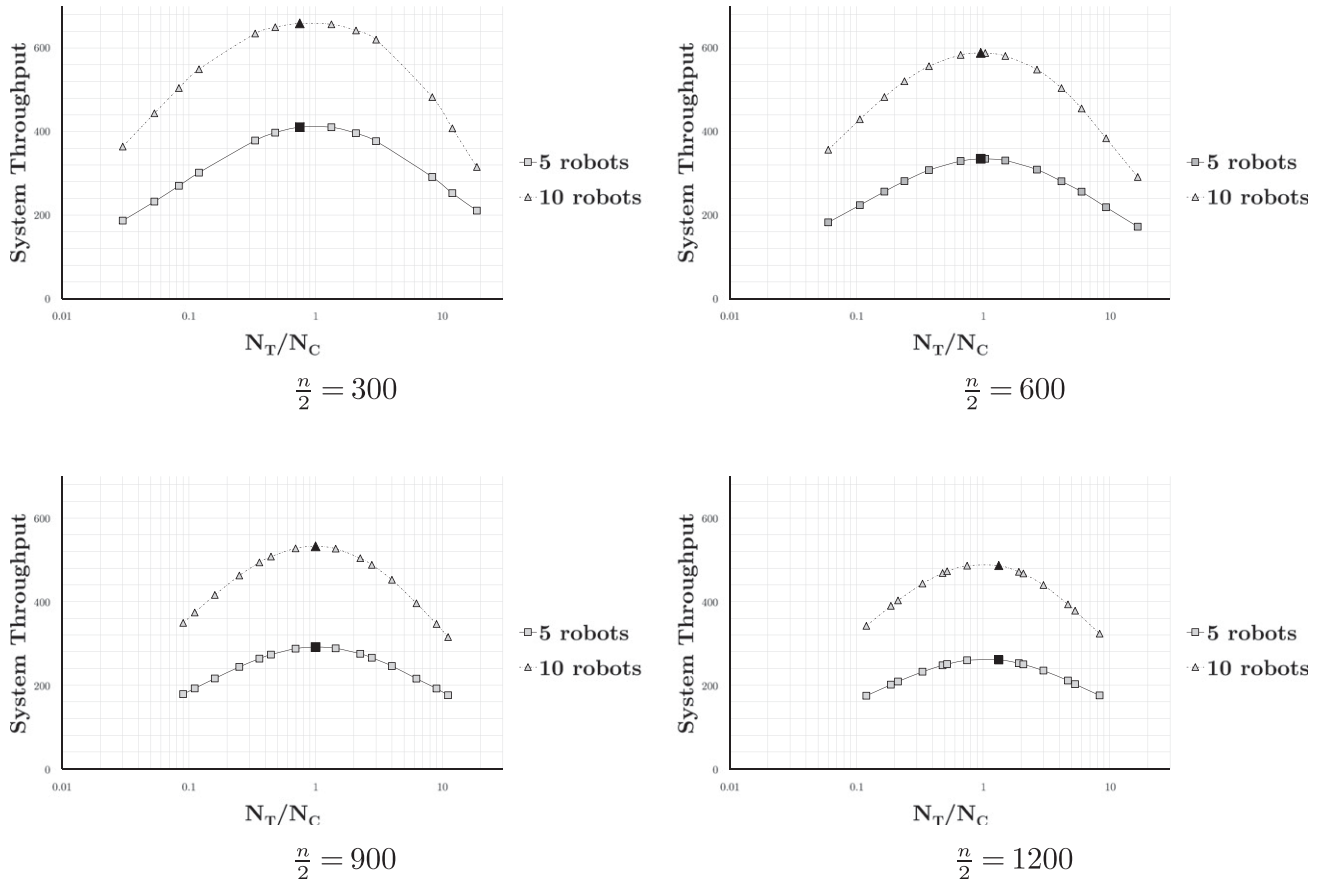


Table 9. Optimal Layout Configurations

$n/2$	R	N_T	N_C	(N_T/N_C)	Throughput (DC/h)
300	5	15	20	0.75	410.54
300	10	15	20	0.75	659.07
600	5	24	25	0.96	334.94
600	10	24	25	0.96	588.35
900	5	30	30	1.00	291.00
900	10	30	30	1.00	532.21
1,200	5	40	30	1.33	260.45
1,200	10	40	30	1.33	486.31

Note. DC/h, dual-command cycles per hour.

cycles per picker per hour). The high picking speed at the L/U point prevents the picking process from becoming the bottleneck. The remaining system parameters are the same as in Table 3. In this scenario, we increase the number of robots in the system from 1 to 15 and observe the system throughput obtained by the two blocking policies. The results are illustrated in Figure 14. The x axis represents the number of robots in the system, and the y axis is the system throughput measured in dual-command cycles per hour.

6.2.1. Analysis and Obtained Insights. The graph (Figure 14) shows that the policies perform approximately the same with a small number of robots in the system (the WOS-0 policy has a slight advantage). However, if we increase the number of robots, the system throughput decreases sharply under the WOS policy. This decrease can be attributed to the waiting robots that block passing robots in the outer loop. The reason for the throughput drop is as follows: there are some waiting positions in the system depending on the number of tiers and rack sections (see Figure 8), and after all of the waiting positions are occupied, the blocking delays cause the system throughput to drop. In our example, the throughput using WOS-0 policy drops if we increase the number of robots from eight to nine, because this can result in a situation in which all four rack sections are full with a robot waiting on top of each. Therefore, the ninth robot will definitely be delayed, resulting in a sudden decrease in the system throughput. Using the WOS-1 policy pushes the point of sharp fall in throughput to a later stage. This phenomenon can be largely mitigated with the REC policy, where the robots circulate in the system while waiting for their rack section to become available. Consequently, they never block other moving robots.

If we compare the two policies, the REC has a lower system throughput, especially when the waiting time is less than the recirculation time and the number of robots is small. However, the REC policy can achieve a higher system throughput without worrying about the blocking delays when the number of robots in the system is increased. Therefore, depending on the number

of robots in the system and the desired throughput level, the company can choose the most suitable policy.

7. Cost-Performance Comparison of the Vertical and Horizontal Systems

Building on the previous sections, we now have the tools to compare the performances of the vertical and horizontal systems. Because the vertical system is primarily used for handling product totes, we also compare its performance with a horizontal system for handling product totes (see Marchet et al. 2012). A horizontal system consists of several aisles with multiple lifts each serving one or multiple aisles. The shuttles are either aisle captive or pooled among aisles. They can also be either tier captive or tier to tier. An L/U point is located at the bottom of each lift. The most comparable horizontal system to the vertical system is a system consisting of several aisles, where each aisle consists of two single-deep storage racks with a dedicated lift and tier-to-tier shuttles. An L/U point is located at one end of each aisle (Figure 15). An extra L/U point can be added at the other end of the aisle, but then, an extra lift needs to be installed as well. The lift can be either discrete or continuous. A discrete lift can only transport one shuttle at a time, whereas the continuous lift (i.e., a sort of vertical conveyor) can transport multiple shuttles simultaneously. However, the discrete lift is faster than a continuous lift. The reason is that a discrete lift only needs to transport one unit at a time, whereas a continuous lift needs to transport many units at a time, which requires much more energy (Marchet et al. 2012, Qimarox BV 2013). It should be noted that the shuttles can reach the first tier without needing a lift. However, a lift is needed to access the remaining tiers for the vertical movement. In this horizontal system, each aisle operates independently. Therefore, for a fair comparison, we analyze the performance of a single aisle of the vertical system with a single aisle of the horizontal system.

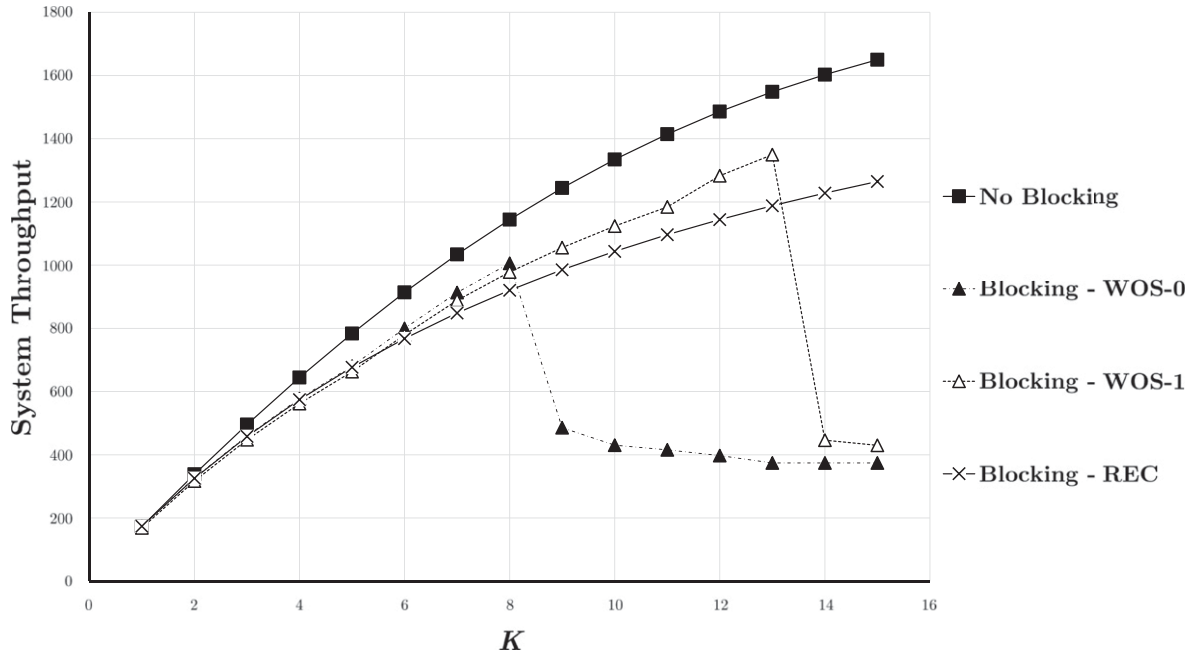
To be in line with the vertical system analysis, we make similar assumptions for analyzing the horizontal system.

Table 10. Optimal Layout Configurations Using Dual-Command Throughput Time Expression Approach

$n/2$	R	N_T	N_C	N_T/N_C	Throughput (DC/h)
300	5	20 (17.92)	15 (16.74)	1.33 (1.07)	410.36
300	10	20 (17.92)	15 (16.74)	1.33 (1.07)	656.83
600	5	25 (25.35)	24 (23.66)	1.04 (1.07)	334.80
600	10	25 (25.35)	24 (23.66)	1.04 (1.07)	588.05
900	5	30 (31.05)	30 (28.99)	1.00 (1.07)	291.00
900	10	30 (31.05)	30 (28.99)	1.00 (1.07)	532.21
1,200	5	40 (35.86)	30 (33.46)	1.33 (1.07)	260.45
1,200	10	40 (35.86)	30 (33.46)	1.33 (1.07)	486.31

Note. DC/h, dual-command cycles per hour.

Figure 14. System Throughput Using Different Blocking Policies



(a) Dual-command cycle orders for shuttles. We assume that all requests are executed in a dual-command cycle.

(b) Same tier. We assume that the dual-command cycle orders are always for a specific tier (i.e., there is always a storage location available in the same tier from which we want to retrieve the new item).

(c) Uniform assignment. We assume that storage and retrieval locations are assigned uniformly in the system.

(d) One shuttle per tier. We assume that only one shuttle can access each tier at a certain time to avoid the risk of having a deadlock in the system.

(e) Constant velocity. We ignore acceleration and deceleration of shuttles and discrete lift during movements in our analysis, and shuttles and lift have a constant velocity.

(f) Dwell point. We assume the point of service completion dwell point policy for the discrete lift. We assume that the shuttles dwell at the L/U point.

In our analysis, we compare the maximum system throughput of the two systems for a given K assuming that there is enough buffer space available in each tier of the horizontal system and each rack section of the vertical system. Then, we calculate the operating cost associated with the system throughput for each system to investigate which system achieves a higher throughput at a lower operating cost. This cost-performance analysis can guide decision makers in choosing the right system based on performance needs and budget constraints. We estimate the performance of the horizontal system by developing a closed queueing network model with unlimited buffer space and estimate the performance of

the vertical system by using the model developed in Section 4.1. The details of the model for the horizontal system are presented in Online Appendix G.

The operating cost is evaluated on the basis of the annualized cost as a sum of three items: equipment cost, floor space cost, and picker cost:

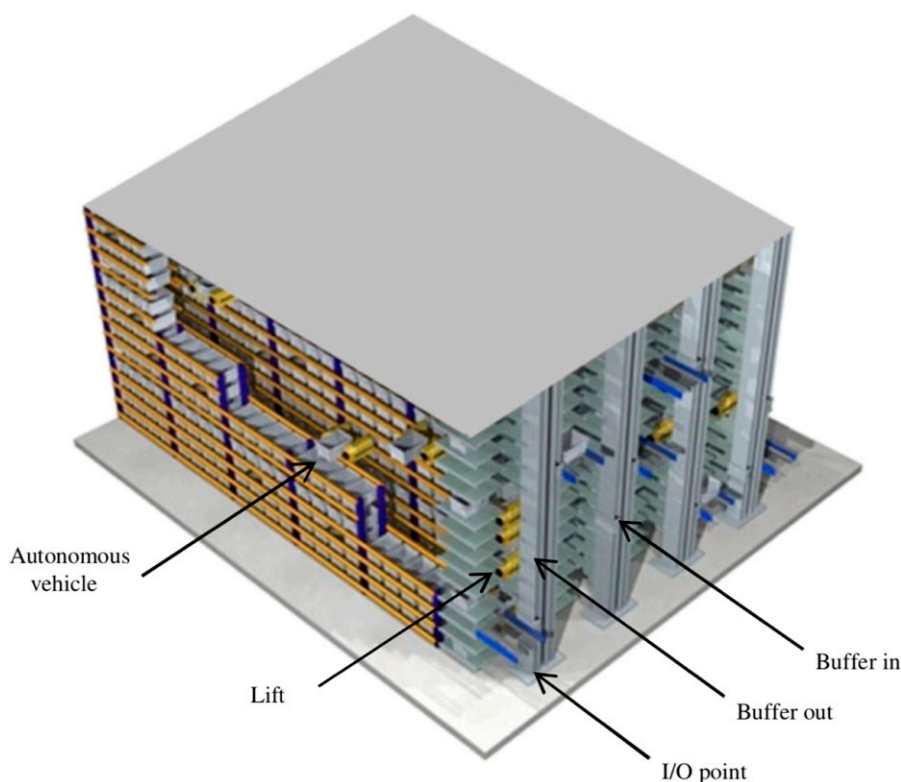
$$Cost = (C_s + C_r)K + C_l N_{LU} + C_A A + C_p N_{LU}, \quad (20)$$

where

- C_s : annualized cost of a shuttle (euros per year),
- C_l : annualized cost of a lift (euros per year),
- C_r : annualized cost of a robot (euros per year),
- C_p : annual cost of a picker (euros per year),
- C_A : annual floor space cost (euros per (meters² year)),
- A : floor space (meters²),
- K : number of robots or shuttles in the system, and
- N_{LU} : number of L/U points in the system.

Note that C_s and C_l are zero for the vertical system and that C_r is zero for the horizontal system. The rack cost is assumed to be the same for both systems, and therefore, it is ignored in our analysis. A is equal to $(w(\text{depth}))N_C$ in both the vertical and horizontal systems. Table 11 shows the parameters of both systems. The shuttles and the discrete lift velocity parameters are taken from MWPVL International (2013). The continuous lift velocity is obtained by analyzing a video of an implementation by Qimarox BV (2013). Different costs are listed in Table 12 (courtesy of Dynamis BV, Marchet et al. (2012), and experts' opinions).

We assume that there is one L/U point available in the systems. We first determine the layout that maximizes the throughput with a given storage capacity

Figure 15. (Color online) Horizontal System (Marchet et al. 2013)

($n/2 = 300, 600, 900, 1200$) given number of robots ($K = 1, \dots, 15$) and a system type. Then, the annualized cost of the configuration is calculated using the costs presented in Table 12 and assuming seven years of service and a 10% interest rate. Figure 16 illustrates the system throughput versus the annualized operating cost.

The vertical system always outperforms the horizontal system with discrete lift (horizontal-d) in terms of both operating costs and system throughput. No matter what the number of storage locations in the system is, the vertical system always has lower operating costs compared with the horizontal-d system for the same throughput. Furthermore, the system throughput in the horizontal-d system saturates as the lift becomes the bottleneck in the system. Therefore, the upper bound for the throughput capacity depends on the capacity of the lift. For instance, when $n/2 = 300$, the throughput capacity cannot be increased to more than 500 dual-command cycles per hour, because the lift becomes the bottleneck at this point. Note that the capacity of the lift depends on the height of the rack,

and because the rack is optimized for every individual instance, increasing the number of robots can lead to a different rack structure with different lift capacity. However, in the vertical system, the system throughput can be increased until the picker becomes the bottleneck at 720 dual-command cycles per hour.

Compared with the horizontal system with continuous lift (horizontal-c), the horizontal-d system performs better when a lower system throughput is required. For a required system throughput of up to about 240 dual-command cycles per hour, the horizontal-d system has lower operating costs, whereas the horizontal-c system has a lower operating cost when a system throughput of more than about 390 dual-command cycles per hour is needed. Between a system throughput of 240 and 390, either system can perform with a lower operating cost depending on the number of storage locations in the system.

The performance of the vertical system also depends on the number of storage locations in the system. When $n/2 = 300$, the vertical system always produces

Table 11. System Parameters

System	$n/2$	w , cm	h , cm	Depth, m	v_r , m/s	v_s , m/s	$v_l(\text{disc})$, m/s	$v_l(\text{cont})$, m/s	τ_l, τ_u , s	$\tau_l, \tau_u(\text{lift})$, s	τ_{LU} , s
Vertical system	300, 600, 900, 1,200	80	32	2.7	1	N/A	N/A	N/A	1.5	N/A	5
Horizontal system	300, 600, 900, 1,200	80	32	2.7	N/A	3.2	3.2	1	2	3	5

Note. N/A, not applicable.

Table 12. Unit Costs

Cost item	Unit of measure	Value	Expected life, y
Shuttle	Euros	15,000	7
Discrete lift	Euros	50,000	7
Continuous lift	Euros	60,000	7
Robot	Euros	15,000	7
Floor space	Euros per (meters ² year)	70	N/A
Picker	Euros per year	35,000	N/A

Note. N/A, not applicable.

a given system throughput at a lower operating cost compared with the horizontal-c system. However, as the number of storage locations increases, the operating cost of the vertical system increases more than the horizontal-c system. As a result, when $n/2 = 1200$, the vertical system has a lower operating cost when the required system throughput is at most 540 dual-command cycles per hour. Beyond this threshold, the horizontal-c system has a lower operating cost.

It is possible to carry out the same analysis assuming that two L/U points are available in each system (at either end of the aisle). The results are presented in Figure 17.

In this situation, the vertical system always has a lower operating cost compared with both horizontal systems. This is because an additional L/U point in the horizontal system requires the installation of an

extra lift, which increases the operating cost significantly. For horizontal-c and horizontal-d systems, similar to the scenario with one L/U point, the operating cost depends on the number of storage locations in the system. When $n/2 = 300$, the horizontal-d system can generate a system throughput of up to 380 dual-command cycles per hour with lower operating cost compared with the horizontal-c system. The number decreases to 560 when $n/2 = 1200$. Similar to the one L/U scenario, the system throughput of the horizontal-d system saturates sooner than the other systems at around 1,060 dual-command cycle per hour, because the lifts become the bottleneck.

Figure 18 summarizes the cost comparison of the three systems. In the figure, the annual operational costs of the vertical system (C_V), the horizontal-c system (C_{h-c}), and the horizontal-d system (C_{h-d}) are compared depending on the number of storage locations and the desired system throughput. Note that the throughput of the horizontal-d system cannot go beyond the black region in each scenario, because the lift becomes at the interface of the black and gray regions.

7.1. Analysis and Obtained Insights.

Except for one scenario, the vertical system produces similar throughput capacity with a lower annualized operating cost compared with the horizontal system. This cost difference can be explained by the fact that

Figure 16. System Throughput vs. Annual Operating Cost—One L/U Point

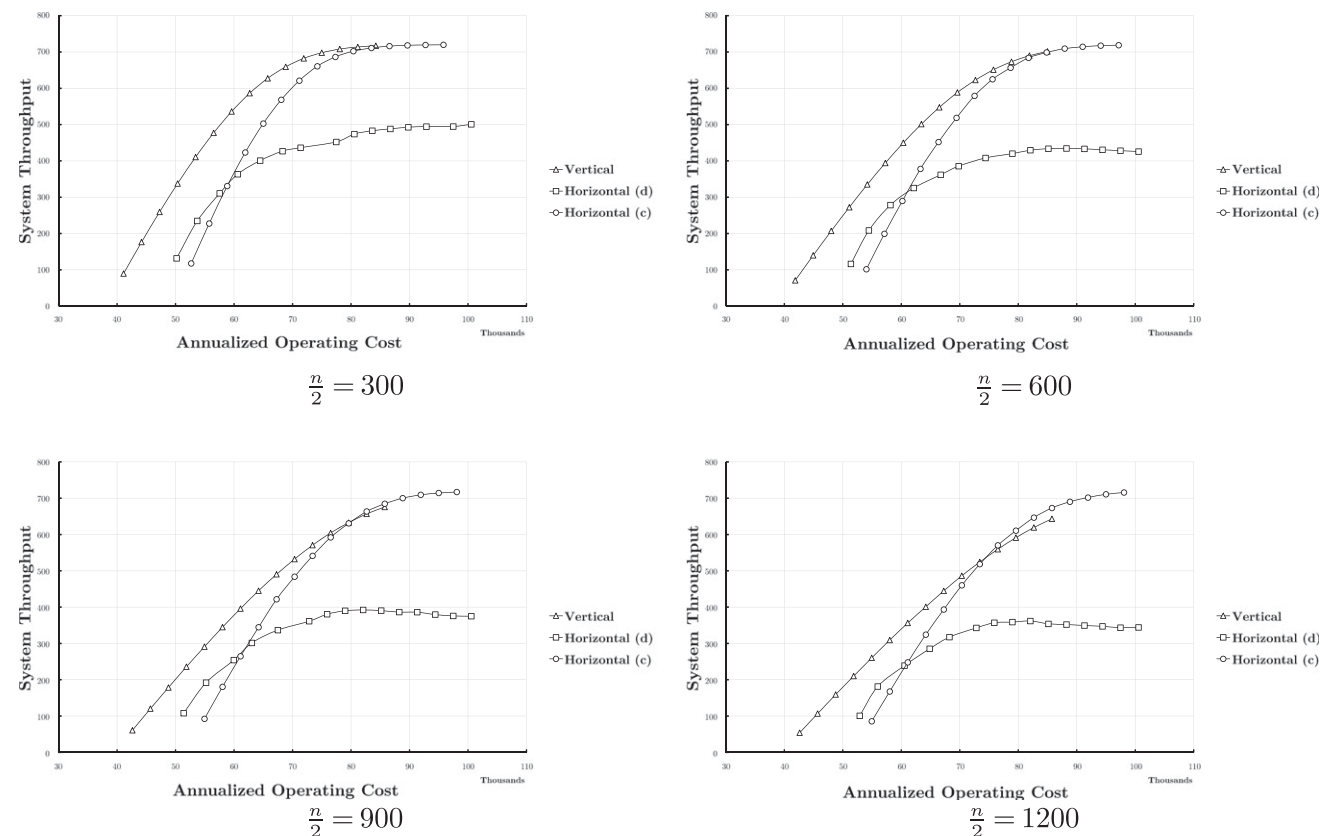
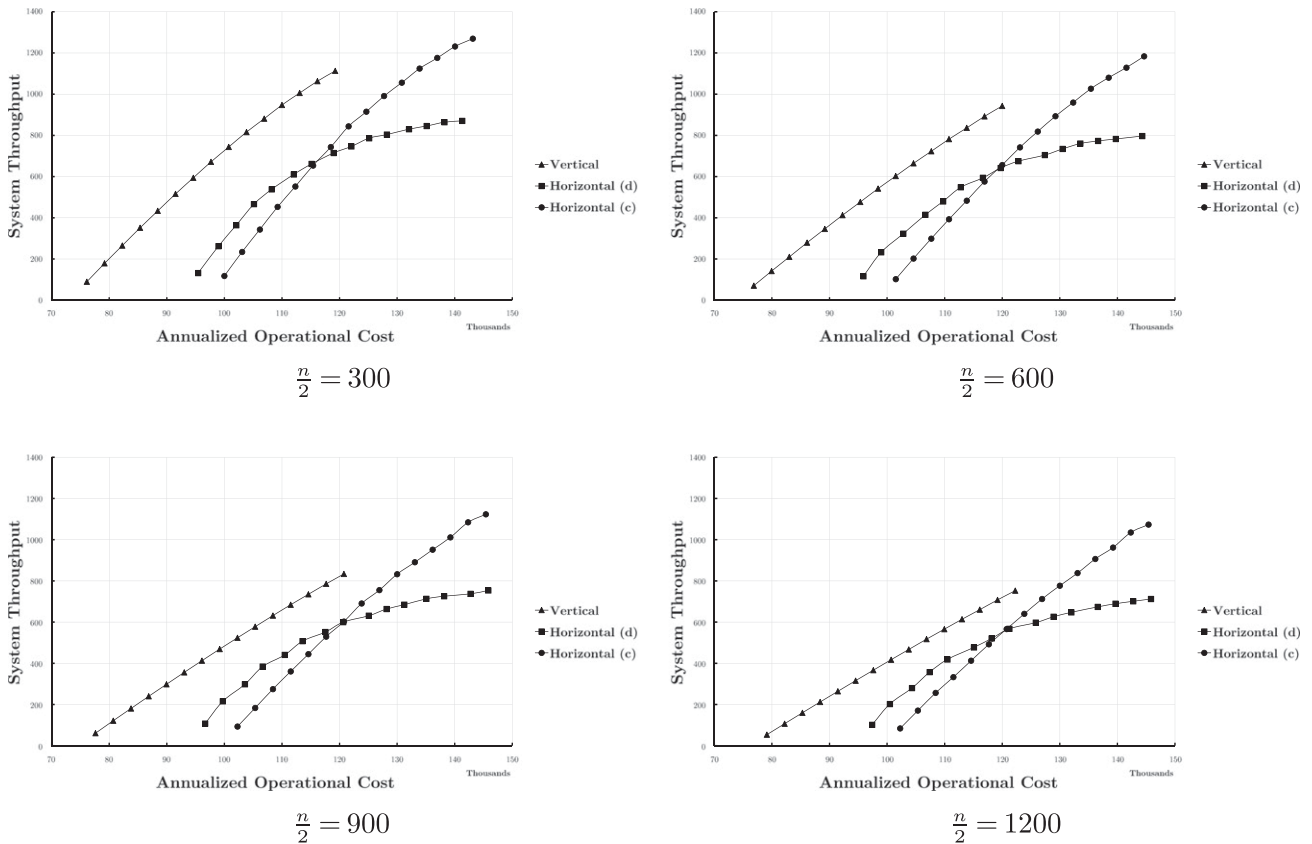


Figure 17. System Throughput vs. Annual Operating Cost—Two L/U Points

the horizontal system uses two material handling resources (lift as well as shuttles), whereas the vertical system only uses the rack climbing robots. The cost difference is even more notable when the systems have two L/U points. Because introducing an additional L/U point in the vertical system does not require the installation of any additional material handling system, its operating cost to achieve a certain throughput capacity is significantly lower even compared with the horizontal system with a continuous lift.

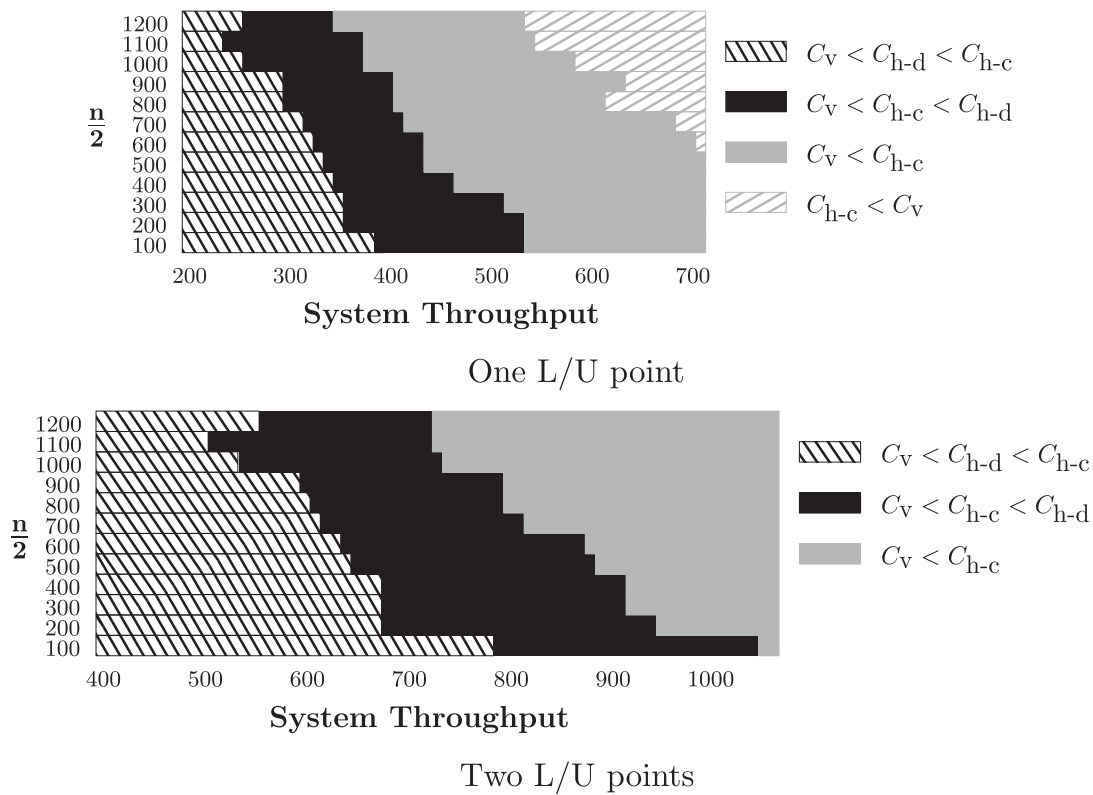
As mentioned earlier, discrete lifts are faster than continuous lift. However, the shuttles can immediately utilize the continuous lifts, whereas they need to wait for the discrete lift. Therefore, when a lower-throughput capacity level is required (which translates to a lower number of shuttles in the system), the shuttles waiting time for the discrete lift is lower than the shuttles travel time with continuous lift. Thus, the horizontal-d system has a better performance with a similar cost. However, as the desired throughput capacity level increases (and consequently, the number of shuttles in the system increases), the shuttles waiting time for the discrete lift will be higher than the shuttle travel time with the continuous lift. Therefore, the horizontal-c system produces better performance at a similar cost.

8. Conclusion and Future Work

In this paper, we study a vertical storage and retrieval system. This is a new robotic-based system that eliminates the need for lifts for the vertical transportation. Robots in this system have the ability to move horizontally and vertically independently and access all of the storage locations within the aisle. We develop analytical models to analyze the performance of the vertical system.

We first assume that there are infinite buffer locations available in each rack section, and therefore, no blocking delays occur in the system. We develop a closed queueing network and estimate the system throughput by using AMVA. We then develop a system dimension optimization model to obtain the optimal system layout for the maximum system throughput. The results show that, when using the provided system parameters, the optimal ratio of height to width (i.e., number of tiers divided by the number of rack sections) is around one. Next, we relax the assumption of having unlimited buffer locations in each rack section to estimate the effect of blocking delays on the system performance. We propose an REC waiting policy for the blocked robots as opposed to the more obvious WOS policy in which the robots wait on top of the occupied

Figure 18. Summary of the Cost Comparison of the Three Systems



rack section until it is unoccupied. We develop two closed queueing networks corresponding to each policy to estimate the performance of the system. We use an approximate method proposed by Akyildiz (1988) to estimate the system throughput in the WOS network. In the case of REC network, we develop a new approximation technique by using the jump-over blocking protocol and an iterative algorithm based on MVA to estimate the performance of the system. This results in a very accurate estimation of the system throughput of the real system. Comparing the results of the two policies, the WOS policy has a slight advantage when the number of robots in the system is small. However, increasing the number of robots results in a sharp decrease in the system throughput, which can be mitigated by adopting the REC policy.

We then compare the performance of the vertical system with the performance of the horizontal system. The horizontal system is modeled by a closed queueing network with an unlimited buffer, and the system throughput is estimated using AMVA. We compare the cost-performance of the vertical system, the horizontal system with a discrete lift, and the horizontal system with a continuous lift. The results indicate that, when there is one L/U point in the system, the vertical system outperforms the horizontal-d system in both operating costs and system throughput. However, compared with the horizontal-c system,

its performance depends on the number of storage locations in the system.

When there are two L/U points in the system, the vertical system always has a lower operating cost compared with both horizontal systems. Horizontal systems require an extra lift when there are two L/U points, which increases operating costs significantly.

One of the unique features of the vertical system is the flexibility of the robots to choose different routing trajectories while processing storage and retrieval transactions. Therefore, it would be interesting for the future studies to find the routing trajectories for the robots that lead to an efficient system design. Furthermore, it would be interesting to examine the effect of different storage policies on the performance of the vertical system. Moreover, it would be interesting to investigate the effect of transaction sequencing policies on system throughput in the presence of multiline orders. In addition, it is possible to also investigate the effect of transaction sequencing on the external transaction waiting time using semiopen queueing network analysis.

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